2024. 금당중3. 1중간

1 ⑤

① -2는 4의 음의 제곱근

② 제곱근 5는 $\sqrt{5}$

③ 정수의 제곱근은

2개, 1개, 0개

④ $\sqrt{9} = 3$.

2 ④

① $\sqrt{2} + \sqrt{3} = \sqrt{2} + \sqrt{3}$

② $\sqrt{3} \times \sqrt{5} = \sqrt{15}$

③ $\sqrt{12} = 2\sqrt{3}$

⑤ $-\sqrt{(-2)^2} = -2$

3 ② 2

$a = 3$, $b = 4$, $c = -5$

$a + b + c = 2$

4 ⑤

5 ① -4

$\overline{AD} = \overline{AB} = \sqrt{5}$

$P : 1 - \sqrt{5}$. $Q : 1 + \sqrt{5}$, $P \times Q = -4$

6 ③ ab^2

$3\sqrt{2} = \sqrt{18} = \sqrt{2} \times \sqrt{3} \times \sqrt{3}$

$= a \times b \times b = ab^2$

7 ① $8\sqrt{3}$

$\sqrt{75} + \frac{3}{\sqrt{3}} + \sqrt{6} \times \sqrt{2}$

$= 5\sqrt{3} + \sqrt{3} + 2\sqrt{3} = 8\sqrt{3}$

8 ③ 1

$m = 10$. $n = \frac{1}{10}$, $mn = 1$

9 ⑤ 8개

$50-x$	x	
0	50	
1	49	
4	46	
9	41	
16	34	
25	25	
36	14	
49	1	8개

10 ② $\boxed{-a-2b}$

$a<0$, $b>0$, $-2a>0$, $a-b<0$

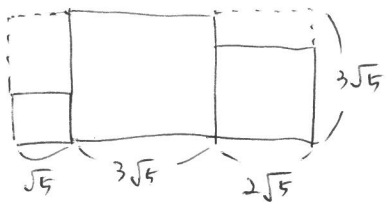
$\sqrt{(-2a)^2} = -2a$

$\sqrt{(a-b)^2} = -a+b$

$\sqrt{(-2a)^2} - (\sqrt{b})^2 - \sqrt{(a-b)^2}$

$= (-2a) - b - (-a+b) = -a-2b$

11 ④ $\boxed{18\sqrt{5}\text{ cm}}$



$2(\sqrt{5}+3\sqrt{5}+2\sqrt{5}+3\sqrt{5}) = 18\sqrt{5}$

12 ④ $\boxed{4}$

$(x+4)(x+8)+k$

$= x^2+12x+32+k$

$\left(\frac{12}{2}\right)^2 = 32+k$, $36 = 32+k$

$k=4$.

13 ③ $\boxed{-5}$

$ab=24$, $a+b=k$

a	b	k
1	24	25
2	12	14
3	8	11
4	6	10
-4	-6	-10
-3	-8	-11
-2	-12	-14
-1	-24	-25

14 ③ $\boxed{20}$

$(\sqrt{5}+3)^2 = 5+6\sqrt{5}+9$

$= 14+6\sqrt{5}$

$a=14$, $b=6$, $a+b=20$

15 ⑤ $\boxed{3x-2y}$

$3x^2-6xy = 3x(x-2y)$

16 ① $\boxed{-2}$

$2x^2-x-3 = (x+1)(2x-3)$

$1x \quad +1$
$2x \quad -3$

$a=1$, $b=-3$, $a+b=-2$

$\boxed{17}$ ② $\boxed{-2\sqrt{3}}$

$x^2y - xy^2 = xy(x-y)$

$= (\sqrt{2}+\sqrt{3})(\sqrt{2}-\sqrt{3})\{(\sqrt{2}+\sqrt{3})-(\sqrt{2}-\sqrt{3})\}$

$= -1 \times 2\sqrt{3} = -2\sqrt{3}$

$\boxed{18}$ ⑤ $\boxed{19}$

$n^2-10n-56 = (n-14)(n+4)$

$n \quad -14$

$n \quad +4$

$n-14=1, \quad n=15$

$(15-14)(15+4) = 19$

$\boxed{19}$ ⑤ $\boxed{155}$

$11\times12\times13\times14+1$

$=(10+1)(10+2)(10+3)(10+4)+1$

$=(10+1)(10+4)(10+2)(10+3)+1$

$=(10^2+5\times10+4)(10^2+5\times10+6)+1$

$=(A+4)(A+6)+1 = A^2+10A+25$

$=(A+5)^2=(10^2+5\times10+5)^2$

$=155^2, \quad \sqrt{155^2}=155$

$\boxed{20}$ ③ $\boxed{\frac{5\sqrt{6}}{2}}$

$x^4-y^4=(x^2+y^2)(x+y)(x-y)$

$x^2+y^2=(x+y)^2-2xy$

$=3-2\times\frac{1}{4}=\frac{5}{2}$

$(x-y)^2=(x+y)^2-4xy$

$2=3-4xy, \quad xy=\frac{1}{4}$

$x^4-y^4=\frac{5}{2}\times\sqrt{3}\times\sqrt{2}=\frac{5\sqrt{6}}{2}$

$\boxed{서11}$ $\boxed{\pm5}$

제곱하여 25가 되는 수

$\boxed{서12}$ $\boxed{3\sqrt{15}-3\sqrt{3}}$

$A=\sqrt{15}+4\sqrt{3}-5\sqrt{3}+2\sqrt{15}$

$=3\sqrt{15}-\sqrt{3}$

$B=\sqrt{3}-3\sqrt{3}=-2\sqrt{3}$

$A+B=3\sqrt{15}-3\sqrt{3}$

$\boxed{서13}$ $\boxed{x-3}$

$4x^2-36=4(x+3)(x-3)$

$2x^2-3x-9=(x-3)(2x+3)$

$\boxed{서14}$ $\boxed{24\sqrt{2}}$

$x=3+2\sqrt{2}, \quad y=3-2\sqrt{2}$

$x^2-y^2=(x+y)(x-y)$

$=6\times4\sqrt{2}=24\sqrt{2}$